Exp: 3b3

23/07/2020

HUB AND SWITCH COMPARISON

AIM:

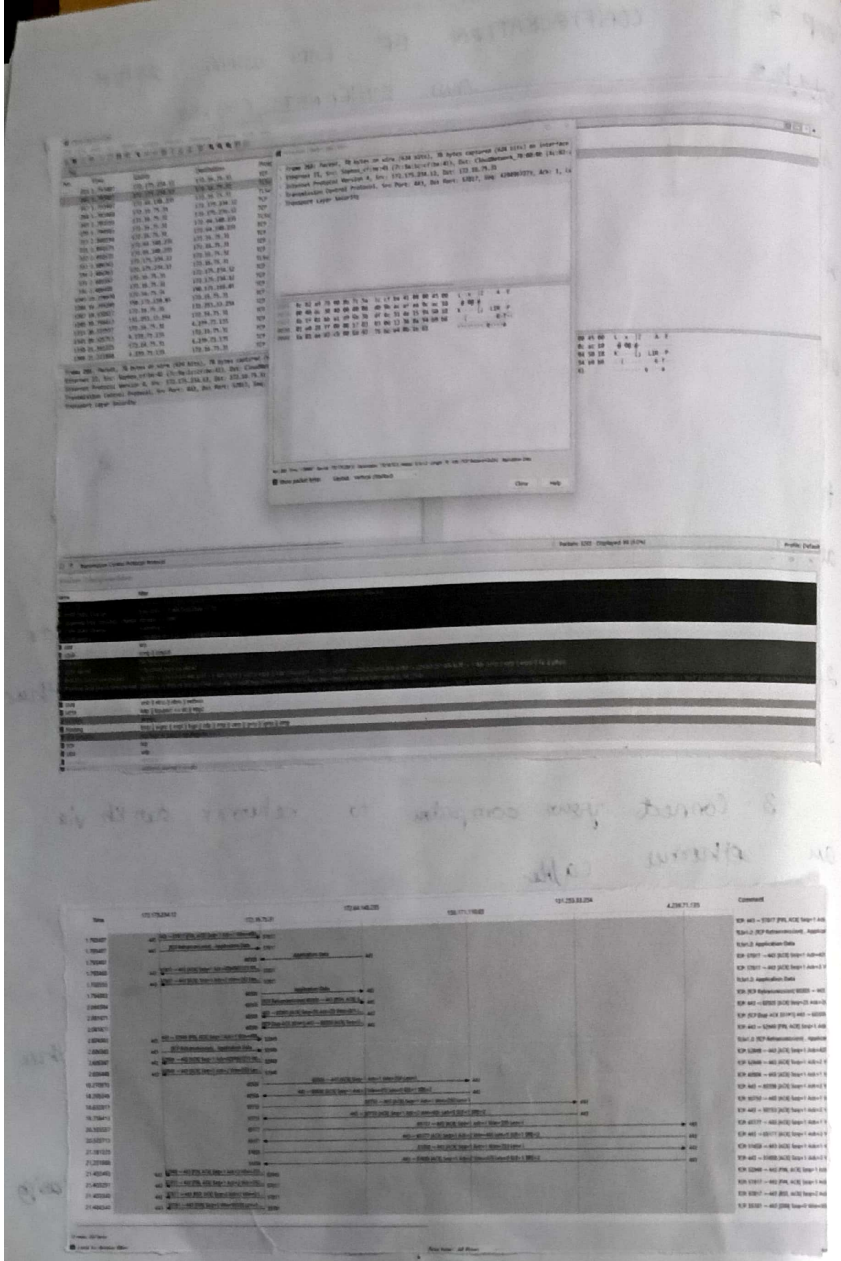
To compare the packet forwarding behaviour of a Hub and a switch using cisco packet tracer

STEPS:

- Create one network using a Hub and another using a switch, each connected to 4 PCs
- Configure IP addresses for all PCs in both networks.
- Send a simple PDU between two PC's in each network.
- Observe Packet Transmission in simulation Mode
- Compare how the Hub & switch forward packets

RESULT:

The Hub broadcasts packet to all connected devices, whereas the switch forwards packets only to the intended destination, Therefore, the switch provides better performance, reduced network traffic, and improved efficiency compared to the HUB



Exp no. 5
30/7/26

EXPERIMENT ON PACKET CAPTURE TOOL - WIRESHARK

AIM:

To understand the feature of wireshark as a packet capture tool and understand encapsulation of information at various layers of a Protocol stack.

PACKET SNIFFER

- Capture Packets sent and received by the Computer
- Display protocol information of Captured packets
- Works passively.

SNIFFER TOOLS:

- TCP dump - Command line Packets capture tool
- Wireshark - Graphical packet analyzer

USES OF NETWORKS:

- Capture network traffic
- Analyzes Protocols

Applications:

- Network troubleshooting
- Network analysis

Capturing Packet:

- Open Wireshark
- Select the network interface
- Click start Capture
- Observe Packet in real time
- Click stop and save the capture

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Exp: 4 30/4/23	CONFIGURATION OF LAN USING SWITCH AND ETHERNET CABLES

AIM:

Setup and configure a LAN using a switch and Ethernet cables in your lab.

STEPS:

1. Plan and design an appropriate network topology taking into account network requirements and equipment location.

2. You can take 4 Computers, a switch with 8, 16 or 24 ports which is sufficient for networks of these sizes

3. Connect your computers to network switch via an ethernet cable.

4. Assign IP address to your PC's

5. Configure a switch network

6. Check the connectivity between switch & other machines

7. Select a folder, go to properties, click sharing tab; share it with everyone on the same LAN

8. Try access the shared folder.

RESULT:

Setup and configuration of a LAN using switch and ethernet cables is successful.

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